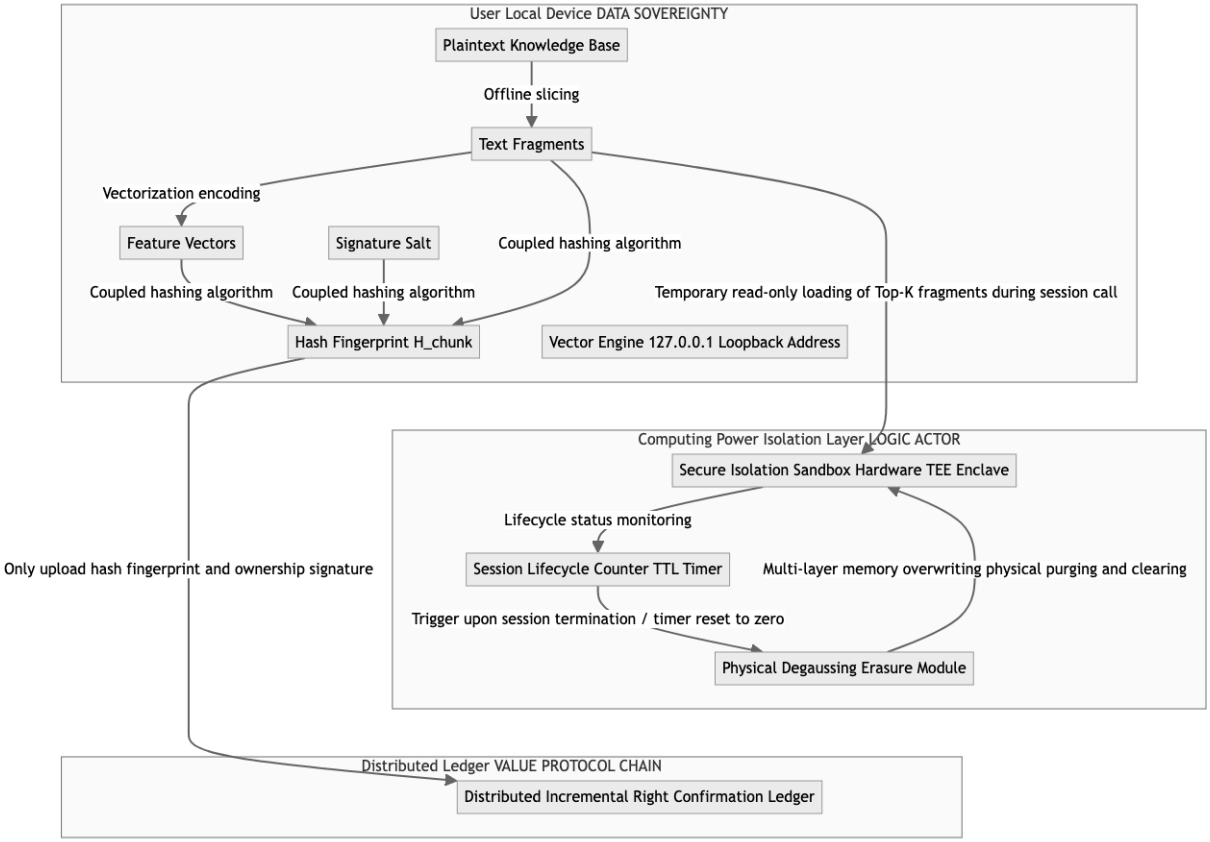

User Local Device DATA SOVEREIGNTY
Plaintext Knowledge Base
Offline slicing
Text Fragments
Vectorization encoding
Feature Vectors
Signature Salt
Coupled hashing algorithm
Coupled hashing algorithm
Coupled hashing algorithm
Temporary read-only loading of Top-K fragments during session call
Hash Fingerprint H_chunk
Vector Engine 127.0.0.1 Loopback Address
Computing Power Isolation Layer LOGIC ACTOR
Secure Isolation Sandbox Hardware TEE Enclave
Lifecycle status monitoring
Session Lifecycle Counter TTL Timer
Multi-layer memory overwriting physical purging and clearing
Only upload hash fingerprint and ownership signature
Trigger upon session termination / timer reset to zero
Physical Degaussing Erasure Module
Distributed Ledger VALUE PROTOCOL CHAIN
Distributed Incremental Right Confirmation Ledger

**图1**

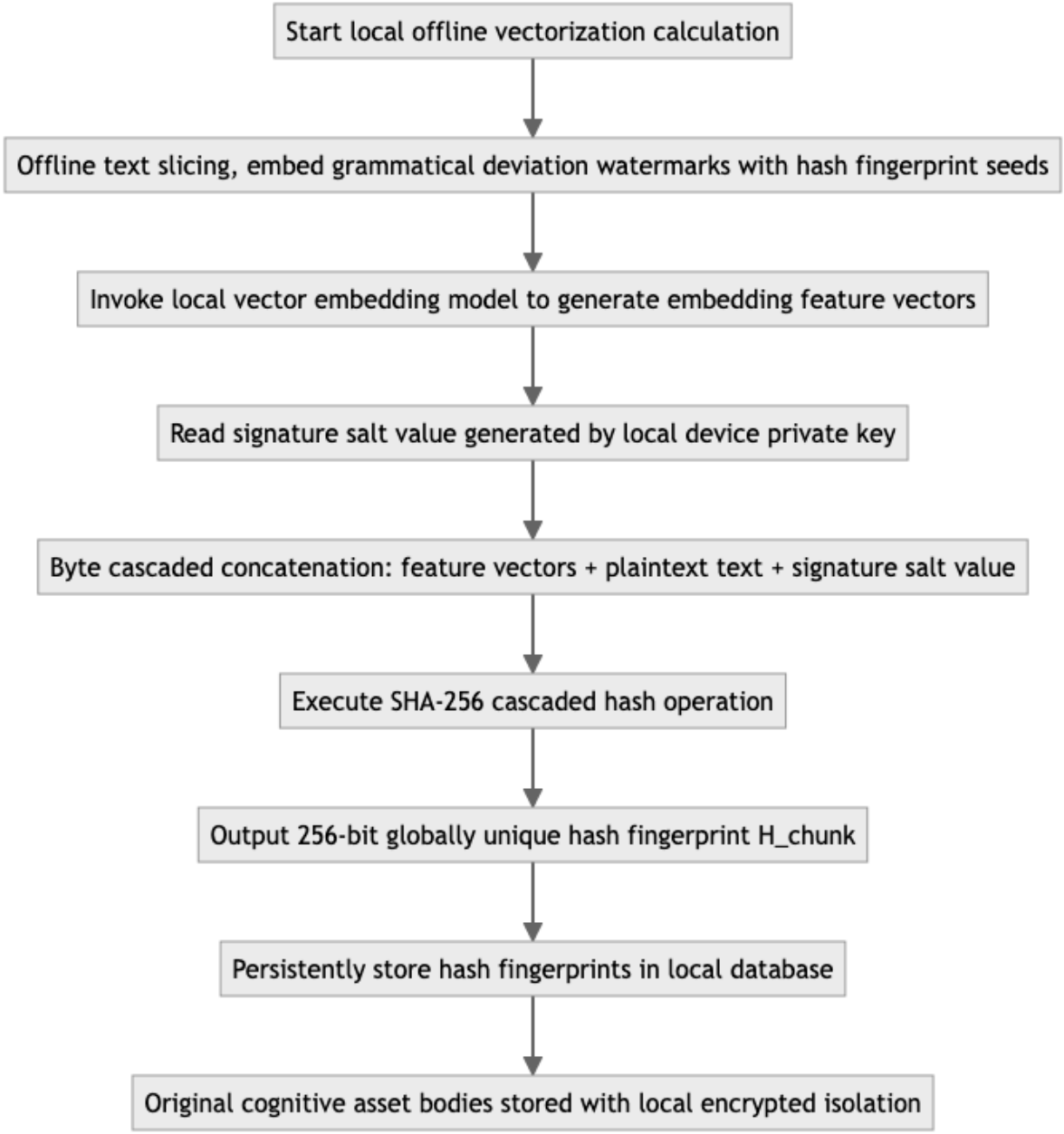

Start local offline vectorization calculation
Offline text slicing, embed grammatical deviation watermarks with hash fingerprint seeds
Invoke local vector embedding model to generate embedding feature vectors
Read signature salt value generated by local device private key
Byte cascaded concatenation: feature vectors + plaintext text + signature salt value
Execute SHA-256 cascaded hash operation
Output 256-bit globally unique hash fingerprint H_chunk
Persistently store hash fingerprints in local database
Original cognitive asset bodies stored with local encrypted isolation

图2

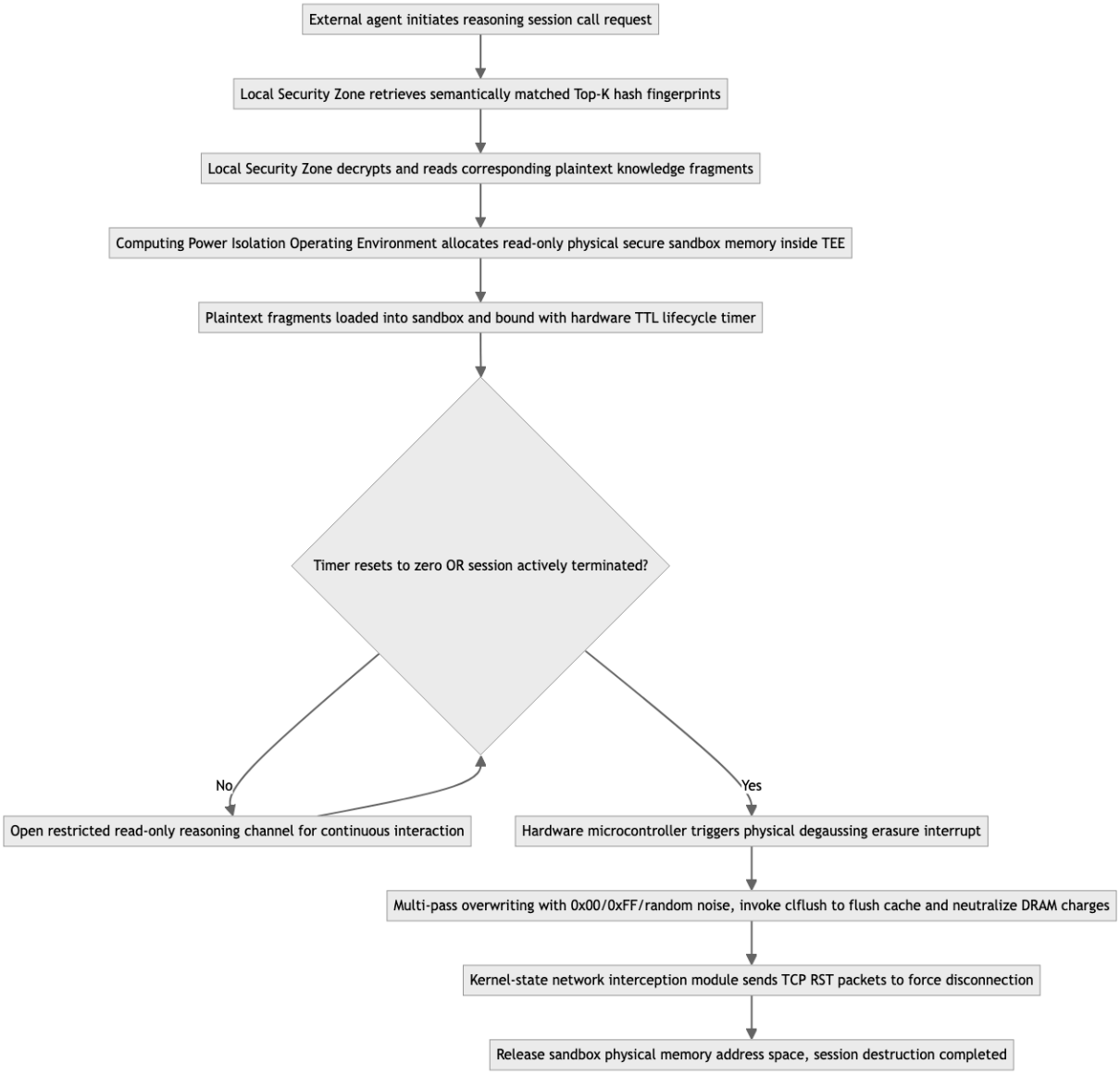

External agent initiates reasoning session call request
Local Security Zone retrieves semantically matched Top-K hash fingerprints
Local Security Zone decrypts and reads corresponding plaintext knowledge fragments
Computing Power Isolation Operating Environment allocates read-only physical secure sandbox memory inside TEE
Plaintext fragments loaded into sandbox and bound with hardware TTL lifecycle timer
Timer resets to zero OR session actively terminated?
No
Yes
Open restricted read-only reasoning channel for continuous interaction
Hardware microcontroller triggers physical degaussing erasure interrupt
Multi-pass overwriting with 0x00/0xFF/random noise, invoke clflush to flush cache and neutralize DRAM charges
Kernel-state network interception module sends TCP RST packets to force disconnection
Release sandbox physical memory address space, session destruction completed

图3